DAY 17 | HANDS-ON ACTIVITIES

DevOps and Cloud Fundamentals: Problem Statements

Seven problems on DevOps and cloud concepts with a local CI simulator. Azure tasks are optional and use Cloud Shell; delete resources afterwards.

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How to work: read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

Problem 1: Map the Loop

Level: Warm-up | **Concepts:** DevOps loop

For your Task Board app from Day 12, write what happens at each stage of the DevOps loop: plan, code, build and test, release, operate and monitor.

Sample run / expected result

```
Plan: user story "filter tasks by priority" on the board
Code: feature branch + pull request
...
```

Hints

- Be concrete: name the tool and the artefact at each stage.

Problem 2: Delivery or Deployment?

Level: Easy | **Concepts:** CI, continuous delivery, continuous deployment

For each team, say which practice they use: (a) tests run on every push, release monthly by hand; (b) every green build waits for a manager's click; (c) every green build goes live.

Sample run / expected result

```
(a) CI only
(b) Continuous delivery
(c) Continuous deployment
```

Hints

- Ask: is there a human approval step?

Problem 3: Choose the Cloud Model

Level: Easy | **Concepts:** IaaS, PaaS, SaaS

Pick a model for: a legacy app needing a custom Windows driver; a FastAPI service; company email; a data science team's notebooks; a database the team does not want to patch.

Sample run / expected result

```
Legacy app: IaaS (VM)
FastAPI: PaaS (App Service)
Email: SaaS
```

Hints

- The more you must control, the further left you go.

Problem 4: Local CI Simulator

Level: Medium | **Concepts:** pipeline steps, fail fast

Using the notebook, run the CI simulator on the sample project. Introduce a failing test, watch the pipeline stop, then fix it and get a green run.

Sample run / expected result

```
[1/3] Syntax check ... OK
[2/3] Unit tests ... FAILED
Pipeline stopped
```

Hints

- Read the failing test output before changing code.

Problem 5: Write a Pipeline YAML

Level: Medium | **Concepts:** azure-pipelines.yml

Write a pipeline that triggers on main, runs on ubuntu-latest, uses Python 3.11, installs requirements, runs flake8 and pytest, and publishes test results.

Sample run / expected result

```
trigger:
  - main
pool:
  vmImage: ubuntu-latest
steps:
  - task: UsePythonVersion@0 ...
```

Hints

- Validate your YAML with yaml.safe_load in the notebook.

Problem 6: Azure Resource Group (optional)

Level: Challenging | **Concepts:** Azure CLI, Cloud Shell

In Azure Cloud Shell, create a resource group with owner and project tags, list it as a table, then delete it. Screenshot each step.

Sample run / expected result

```
az group create -n rg-day17-yourname -l centralindia --tags owner=you project=training
az group list -o table
az group delete -n rg-day17-yourname
```

Hints

- Use your own subscription only with your organisation's permission.
- Always delete test resource groups.

Problem 7: Boards Backlog

Level: Stretch | **Concepts:** Azure Boards, user stories

Write 6 user stories for the Task Board with acceptance criteria, group them into 2 sprints, and mark one as a bug. If you have Azure DevOps access, enter them in Boards.

Sample run / expected result

```
As a user, I want to filter tasks by priority so that I can focus on urgent work.
Acceptance: filter shows only matching tasks; clearing it shows all.
```

Hints

- Use the format: As a ..., I want ..., so that ...

Ganbatte (頑張って): do your best